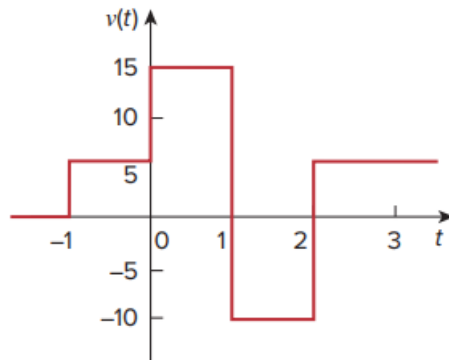


Problem

Express $v(t)$ in Fig. 7.105 in terms of step functions.

Fig. 7.105:



Step-by-step solution

Step 1 of 1

Refer to Figure 7.105 in the textbook for a signal $v(t)$.

Express the signal $v(t)$ in terms of step functions.

The general representation of a unit step function is,

$$u(t) = \begin{cases} 1, & t > 0, \\ 0, & t \leq 0. \end{cases}$$

The signal $v(t)$ has four sections of step functions.

From $-1 \leq t < 0$, the signal has 5 units of increasing amplitude. Therefore, the expression for the signal $v(t)$ in this limit is,

$$v(t) = 5u(t+1)$$

From $0 \leq t < 1$, the signal has 10 units of increasing amplitude starting from 5. Therefore, the expression for the signal $v(t)$ in this limit is,

$$v(t) = 10u(t)$$

From $1 \leq t < 2$, the signal has 25 units of decreasing amplitude starting from 15. Therefore, the expression for the signal $v(t)$ in this limit is,

$$v(t) = -25u(t-1)$$

From $t \geq 2$, the signal has 15 units of increasing amplitude starting from -10 . Therefore, the expression for the signal $v(t)$ in this limit is,

$$v(t) = 15u(t-2)$$

All these signals are added to get the final signal $v(t)$.

Therefore, the expression for the $v(t)$ in terms of step functions is,

$$v(t) = 5u(t+1) + 10u(t) - 25u(t-1) + 15u(t-2)$$